Complete Kinematic Equations for Spatial 7R Mechanism

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1 Overview

This document outlines the complete set of closed-loop equations derived using the Loop-Splitting method. The mechanism parameters are normalized using a base length scale L.

Notation:

- L: Base length unit (e.g., 40mm).
- θ_i: Joint angle for joint i.
- Equations are derived from $T_{LHS} = T_{RHS}$.

2 Permutation 1: $T_1T_2T_3T_4 = T_7^{-1}T_6^{-1}T_5^{-1}$

$$
\begin{aligned}
(n_x): &\quad -\sin(\theta_4)\sin(\theta_1+\theta_2) - \cos(\theta_3)\cos(\theta_4)\cos(\theta_1+\theta_2) && (1)\\
&= -\sin(\theta_5)\sin(\theta_7) - \cos(\theta_5)\cos(\theta_6)\cos(\theta_7) && (2)\\
[8pt](o_x): &\quad \sin(\theta_3)\cos(\theta_1+\theta_2) && (3)\\
&= -\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5) && (4)\\
[8pt](a_x): &\quad \sin(\theta_4)\cos(\theta_3)\cos(\theta_1+\theta_2) - \sin(\theta_1+\theta_2)\cos(\theta_4) && (5)\\
&= -\sin(\theta_6)\cos(\theta_7) && (6)\\
[8pt](p_x): &\quad L\left(-2.5\sin(\theta_4)\sin(\theta_1+\theta_2) + 1.0\cos(\theta_1) - 2.5\cos(\theta_3)\cos(\theta_4)\cos(\theta_1+\theta_2) + 2.0\cos(\theta_1+\theta_2)\right) && (7)\\
&= L\left(2.5\sin(\theta_7) + 2.5\cos(\theta_6)\cos(\theta_7) - 2.0\right) && (8)\\
[8pt](n_y): &\quad \sin(\theta_4)\cos(\theta_1+\theta_2) - \sin(\theta_1+\theta_2)\cos(\theta_3)\cos(\theta_4) && (9)\\
&= -\sin(\theta_6)\cos(\theta_5) && (10)\\
[8pt](o_y): &\quad \sin(\theta_3)\sin(\theta_1+\theta_2) && (11)\\
&= -\sin(\theta_5)\sin(\theta_6) && (12)\\
[8pt](a_y): &\quad \sin(\theta_4)\sin(\theta_1+\theta_2)\cos(\theta_3) + \cos(\theta_4)\cos(\theta_1+\theta_2) && (13)\\
&= \cos(\theta_6) && (14)\\
[8pt](p_y): &\quad L\left(1.0\sin(\theta_1) + 2.5\sin(\theta_4)\cos(\theta_1+\theta_2) - 2.5\sin(\theta_1+\theta_2)\cos(\theta_3)\cos(\theta_4) + 2.0\sin(\theta_1+\theta_2)\right) && (15)\\
&= 2.5L\sin(\theta_6) && (16)\\
[8pt](n_z): &\quad \sin(\theta_3)\cos(\theta_4) && (17)\\
&= -\sin(\theta_5)\cos(\theta_7) + \sin(\theta_7)\cos(\theta_5)\cos(\theta_6) && (18)\\
[8pt](o_z): &\quad \cos(\theta_3) && (19)\\
&= \sin(\theta_5)\sin(\theta_7)\cos(\theta_6) + \cos(\theta_5)\cos(\theta_7) && (20)\\
[8pt](a_z): &\quad -\sin(\theta_3)\sin(\theta_4) && (21)\\
&= \sin(\theta_6)\sin(\theta_7) && (22)\\
[8pt](p_z): &\quad L\left(2.5\sin(\theta_3)\cos(\theta_4) + 5.0\right) && (23)\\
&= -2.5L\left(\sin(\theta_7)\cos(\theta_6) - \cos(\theta_7)\right) && (24)\\
[8pt] & && (25)
\end{aligned}
$$

3 Permutation 2: $T_2T_3T_4T_5 = T_1^{-1}T_7^{-1}T_6^{-1}$

$$(n_x): \quad -(\sin(\theta_2)\sin(\theta_4)+\cos(\theta_2)\cos(\theta_3)\cos(\theta_4))\cos(\theta_5)+\sin(\theta_3)\sin(\theta_5)\cos(\theta_2) \tag{26}$$

$$= -\sin(\theta_1)\sin(\theta_6)-\cos(\theta_1)\cos(\theta_6)\cos(\theta_7) \tag{27}$$

$$[8pt](o_x): \quad -\sin(\theta_2)\cos(\theta_4)+\sin(\theta_4)\cos(\theta_2)\cos(\theta_3) \tag{28}$$

$$= \sin(\theta_1)\cos(\theta_6)-\sin(\theta_6)\cos(\theta_1)\cos(\theta_7) \tag{29}$$

$$[8pt](a_x): \quad -(\sin(\theta_2)\sin(\theta_4)+\cos(\theta_2)\cos(\theta_3)\cos(\theta_4))\sin(\theta_5)-\sin(\theta_3)\cos(\theta_2)\cos(\theta_5) \tag{30}$$

$$= -\sin(\theta_7)\cos(\theta_1) \tag{31}$$

$$[8pt](p_x): \quad L(-2.5(\sin(\theta_2)\sin(\theta_4)+\cos(\theta_2)\cos(\theta_3)\cos(\theta_4))\cos(\theta_5)-2.5\sin(\theta_2)\sin(\theta_4)+2.5\sin(\theta_3)\sin \tag{32}$$

$$= L(2.5\sin(\theta_7)\cos(\theta_1)-2.0\cos(\theta_1)-1.0) \tag{33}$$

$$[8pt](n_y): \quad (-\sin(\theta_2)\cos(\theta_3)\cos(\theta_4)+\sin(\theta_4)\cos(\theta_2))\cos(\theta_5)+\sin(\theta_2)\sin(\theta_3)\sin(\theta_5) \tag{34}$$

$$= \sin(\theta_1)\cos(\theta_6)\cos(\theta_7)-\sin(\theta_6)\cos(\theta_1) \tag{35}$$

$$[8pt](o_y): \quad \sin(\theta_2)\sin(\theta_4)\cos(\theta_3)+\cos(\theta_2)\cos(\theta_4) \tag{36}$$

$$= \sin(\theta_1)\sin(\theta_6)\cos(\theta_7)+\cos(\theta_1)\cos(\theta_6) \tag{37}$$

$$[8pt](a_y): \quad (-\sin(\theta_2)\cos(\theta_3)\cos(\theta_4)+\sin(\theta_4)\cos(\theta_2))\sin(\theta_5)-\sin(\theta_2)\sin(\theta_3)\cos(\theta_5) \tag{38}$$

$$= \sin(\theta_1)\sin(\theta_7) \tag{39}$$

$$[8pt](p_y): \quad L(2.5(-\sin(\theta_2)\cos(\theta_3)\cos(\theta_4)+\sin(\theta_4)\cos(\theta_2))\cos(\theta_5)+2.5\sin(\theta_2)\sin(\theta_3)\sin(\theta_5)-2.5\si \tag{40}$$

$$= L(2.0-2.5\sin(\theta_7))\sin(\theta_1) \tag{41}$$

$$[8pt](n_z): \quad \sin(\theta_3)\cos(\theta_4)\cos(\theta_5)+\sin(\theta_5)\cos(\theta_3) \tag{42}$$

$$= \sin(\theta_7)\cos(\theta_6) \tag{43}$$

$$[8pt](o_z): \quad -\sin(\theta_3)\sin(\theta_4) \tag{44}$$

$$= \sin(\theta_6)\sin(\theta_7) \tag{45}$$

$$[8pt](a_z): \quad \sin(\theta_3)\sin(\theta_5)\cos(\theta_4)-\cos(\theta_3)\cos(\theta_5) \tag{46}$$

$$= -\cos(\theta_7) \tag{47}$$

$$[8pt](p_z): \quad L(2.5\sin(\theta_3)\cos(\theta_4)\cos(\theta_5)+2.5\sin(\theta_3)\cos(\theta_4)+2.5\sin(\theta_5)\cos(\theta_3)+5.0) \tag{48}$$

$$= 2.5L\cos(\theta_7) \tag{49}$$

$$[8pt] \tag{50}$$

4 Permutation 3: $T_3T_4T_5T_6 = T_2^{-1}T_1^{-1}T_7^{-1}$

$$(n_x): \quad (-\sin(\theta_3)\sin(\theta_5) + \cos(\theta_3)\cos(\theta_4)\cos(\theta_5))\cos(\theta_6) - \sin(\theta_4)\sin(\theta_6)\cos(\theta_3) \tag{51}$$

$$= \frac{(\cos(2\theta_1) + \cos(2\theta_2) + \cos(2\theta_1 + 2\theta_2) + 1)\cos(\theta_7)}{4\cos(\theta_1)\cos(\theta_2)} \tag{52}$$

$$[8pt](o_x): \quad \sin(\theta_3)\cos(\theta_5) + \sin(\theta_5)\cos(\theta_3)\cos(\theta_4) \tag{53}$$

$$= \frac{(\cos(2\theta_1) + \cos(2\theta_2) + \cos(2\theta_1 + 2\theta_2) + 1)\sin(\theta_7)}{4\cos(\theta_1)\cos(\theta_2)} \tag{54}$$

$$[8pt](a_x): \quad (\sin(\theta_3)\sin(\theta_5) - \cos(\theta_3)\cos(\theta_4)\cos(\theta_5))\sin(\theta_6) - \sin(\theta_4)\cos(\theta_3)\cos(\theta_6) \tag{55}$$

$$= -\frac{\sin(\theta_1) + \sin(\theta_1 + 2\theta_2)}{2\cos(\theta_2)} \tag{56}$$

$$[8pt](p_x): \quad -2.5L\left(\sqrt{2}\sin(\theta_3)\cos\left(\theta_5 + \frac{\pi}{4}\right) + \sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_3)\cos(\theta_4) + \cos(\theta_3)\cos(\theta_4)\right) \tag{57}$$

$$= \frac{L(-2.0\cos^2(\theta_1)\cos^2(\theta_2) - 1.0\cos(\theta_1)\cos^2(\theta_2) - 2.0\cos(\theta_1)\cos(\theta_2) + 0.25\cos(2\theta_1 - 2\theta_2) -}{\cos(\theta_1)\cos(\theta_2)} \tag{58}$$

$$[8pt](n_y): \quad (\sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + \sin(\theta_5)\cos(\theta_3))\cos(\theta_6) - \sin(\theta_3)\sin(\theta_4)\sin(\theta_6) \tag{59}$$

$$= \sin(\theta_7) \tag{60}$$

$$[8pt](o_y): \quad \sin(\theta_3)\sin(\theta_5)\cos(\theta_4) - \cos(\theta_3)\cos(\theta_5) \tag{61}$$

$$= -\cos(\theta_7) \tag{62}$$

$$[8pt](a_y): \quad -(\sin(\theta_3)\cos(\theta_4)\cos(\theta_5) + \sin(\theta_5)\cos(\theta_3))\sin(\theta_6) - \sin(\theta_3)\sin(\theta_4)\cos(\theta_6) \tag{63}$$

$$= 0 \tag{64}$$

$$[8pt](p_y): \quad 2.5L\left(-\sqrt{2}\sin(\theta_3)\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_4) - \sin(\theta_3)\cos(\theta_4) + \sqrt{2}\cos(\theta_3)\cos\left(\theta_5 + \frac{\pi}{4}\right)\right) \tag{65}$$

$$= 5.0L \tag{66}$$

$$[8pt](n_z): \quad -\sin(\theta_4)\cos(\theta_5)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_4) \tag{67}$$

$$= -\frac{(\sin(\theta_2) + \sin(2\theta_1 + \theta_2))\cos(\theta_7)}{2\cos(\theta_1)} \tag{68}$$

$$[8pt](o_z): \quad -\sin(\theta_4)\sin(\theta_5) \tag{69}$$

$$= -\frac{(\sin(\theta_2) + \sin(2\theta_1 + \theta_2))\sin(\theta_7)}{2\cos(\theta_1)} \tag{70}$$

$$[8pt](a_z): \quad \sin(\theta_4)\sin(\theta_6)\cos(\theta_5) - \cos(\theta_4)\cos(\theta_6) \tag{71}$$

$$= -\cos(\theta_1 + \theta_2) \tag{72}$$

$$[8pt](p_z): \quad 2.5L\left(\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right) + 1\right)\sin(\theta_4) \tag{73}$$

$$= L(2.0\sin(\theta_1)\cos(\theta_2) + 2.0\sin(\theta_2)\cos(\theta_1) + 1.0\sin(\theta_2)) \tag{74}$$

$$[8pt] \tag{75}$$

5 Permutation 4: $T_4T_5T_6T_7 = T_3^{-1}T_2^{-1}T_1^{-1}$

$$(n_x): \quad (\sin(\theta_4)\sin(\theta_6) - \cos(\theta_4)\cos(\theta_5)\cos(\theta_6))\cos(\theta_7) - \sin(\theta_5)\sin(\theta_7)\cos(\theta_4) \tag{76}$$

$$= -\frac{(\cos(2\theta_1) + \cos(2\theta_2) + \cos(2\theta_1 + 2\theta_2) + 1)\cos(\theta_3)}{4\cos(\theta_1)\cos(\theta_2)} \tag{77}$$

$$[8pt](o_x): \quad -\sin(\theta_4)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_4)\cos(\theta_5) \tag{78}$$

$$= -\frac{(\sin(\theta_1) + \sin(\theta_1 + 2\theta_2))\cos(\theta_3)}{2\cos(\theta_2)} \tag{79}$$

$$[8pt](a_x): \quad (-\sin(\theta_4)\sin(\theta_6) + \cos(\theta_4)\cos(\theta_5)\cos(\theta_6))\sin(\theta_7) - \sin(\theta_5)\cos(\theta_4)\cos(\theta_7) \tag{80}$$

$$= \sin(\theta_3) \tag{81}$$

$$[8pt](p_x): \quad L\left(2.0\sin(\theta_4)\sin(\theta_6)\cos(\theta_7) - 2.0\sin(\theta_5)\sin(\theta_7)\cos(\theta_4) + 2.5\sqrt{2}\sin\left(\theta_5 + \frac{\pi}{4}\right)\cos(\theta_4) - 2.0 \tag{82}$$

$$= \frac{L(1.25\sin(\theta_2 - 2\theta_3) - 1.25\sin(\theta_2 + 2\theta_3) + 1.0\cos^2(\theta_2)\cos^2(\theta_3) + 2.0\cos(\theta_2)\cos^2(\theta_3))}{\cos(\theta_2)\cos(\theta_3)} \tag{83}$$

$$[8pt](n_y): \quad -(\sin(\theta_4)\cos(\theta_5)\cos(\theta_6) + \sin(\theta_6)\cos(\theta_4))\cos(\theta_7) - \sin(\theta_4)\sin(\theta_5)\sin(\theta_7) \tag{84}$$

$$= -\frac{\sin(\theta_2) + \sin(2\theta_1 + \theta_2)}{2\cos(\theta_1)} \tag{85}$$

$$[8pt](o_y): \quad -\sin(\theta_4)\sin(\theta_6)\cos(\theta_5) + \cos(\theta_4)\cos(\theta_6) \tag{86}$$

$$= \cos(\theta_1 + \theta_2) \tag{87}$$

$$[8pt](a_y): \quad (\sin(\theta_4)\cos(\theta_5)\cos(\theta_6) + \sin(\theta_6)\cos(\theta_4))\sin(\theta_7) - \sin(\theta_4)\sin(\theta_5)\cos(\theta_7) \tag{88}$$

$$= 0 \tag{89}$$

$$[8pt](p_y): \quad L\left(-2.0\sin(\theta_4)\sin(\theta_5)\sin(\theta_7) + 2.5\sqrt{2}\sin(\theta_4)\sin\left(\theta_5 + \frac{\pi}{4}\right) - 2.0\sin(\theta_4)\cos(\theta_5)\cos(\theta_6)\cos \tag{90}$$

$$= 1.0L\sin(\theta_2) \tag{91}$$

$$[8pt](n_z): \quad \sin(\theta_5)\cos(\theta_6)\cos(\theta_7) - \sin(\theta_7)\cos(\theta_5) \tag{92}$$

$$= -\frac{(\cos(2\theta_1) + \cos(2\theta_2) + \cos(2\theta_1 + 2\theta_2) + 1)\sin(\theta_3)}{4\cos(\theta_1)\cos(\theta_2)} \tag{93}$$

$$[8pt](o_z): \quad \sin(\theta_5)\sin(\theta_6) \tag{94}$$

$$= -\frac{(\sin(\theta_1) + \sin(\theta_1 + 2\theta_2))\sin(\theta_3)}{2\cos(\theta_2)} \tag{95}$$

$$[8pt](a_z): \quad -\sin(\theta_5)\sin(\theta_7)\cos(\theta_6) - \cos(\theta_5)\cos(\theta_7) \tag{96}$$

$$= \cos(\theta_3 + 1.0\pi) \tag{97}$$

$$[8pt](p_z): \quad L\left(2.0\sin(\theta_5)\cos(\theta_6)\cos(\theta_7) - 2.0\sin(\theta_7)\cos(\theta_5) + 2.5\sqrt{2}\cos\left(\theta_5 + \frac{\pi}{4}\right)\right) \tag{98}$$

$$= L(1.0\sin(\theta_3)\cos(\theta_2) + 2.0\sin(\theta_3) + 5.0\cos(\theta_3)) \tag{99}$$

$$[8pt] \tag{100}$$

6 Permutation 5: $T_5T_6T_7T_1 = T_4^{-1}T_3^{-1}T_2^{-1}$

$$(n_x): \quad -(\sin(\theta_5)\sin(\theta_7)+\cos(\theta_5)\cos(\theta_6)\cos(\theta_7))\cos(\theta_1)-\sin(\theta_1)\sin(\theta_6)\cos(\theta_5) \tag{101}$$

$$= -\sin(\theta_2)\sin(\theta_4)-\cos(\theta_2)\cos(\theta_3)\cos(\theta_4) \tag{102}$$

$$[8pt](o_x): \quad (\sin(\theta_5)\sin(\theta_7)+\cos(\theta_5)\cos(\theta_6)\cos(\theta_7))\sin(\theta_1)-\sin(\theta_6)\cos(\theta_1)\cos(\theta_5) \tag{103}$$

$$= -\sin(\theta_2)\cos(\theta_3)\cos(\theta_4)+\sin(\theta_4)\cos(\theta_2) \tag{104}$$

$$[8pt](a_x): \quad -\sin(\theta_5)\cos(\theta_7)+\sin(\theta_7)\cos(\theta_5)\cos(\theta_6) \tag{105}$$

$$= \sin(\theta_3)\cos(\theta_4) \tag{106}$$

$$[8pt](p_x): \quad L(-1.0(\sin(\theta_5)\sin(\theta_7)+\cos(\theta_5)\cos(\theta_6)\cos(\theta_7))\cos(\theta_1)-1.0\sin(\theta_1)\sin(\theta_6)\cos(\theta_5)-2.0\text{s} \tag{107}$$

$$= -L(5.0\sin(\theta_3)\cos(\theta_4)-2.0\cos(\theta_3)\cos(\theta_4)+2.5) \tag{108}$$

$$[8pt](n_y): \quad (-\sin(\theta_5)\cos(\theta_6)\cos(\theta_7)+\sin(\theta_7)\cos(\theta_5))\cos(\theta_1)-\sin(\theta_1)\sin(\theta_5)\sin(\theta_6) \tag{109}$$

$$= \sin(\theta_3)\cos(\theta_2) \tag{110}$$

$$[8pt](o_y): \quad (\sin(\theta_5)\cos(\theta_6)\cos(\theta_7)-\sin(\theta_7)\cos(\theta_5))\sin(\theta_1)-\sin(\theta_5)\sin(\theta_6)\cos(\theta_1) \tag{111}$$

$$= \sin(\theta_2)\sin(\theta_3) \tag{112}$$

$$[8pt](a_y): \quad \sin(\theta_5)\sin(\theta_7)\cos(\theta_6)+\cos(\theta_5)\cos(\theta_7) \tag{113}$$

$$= \cos(\theta_3) \tag{114}$$

$$[8pt](p_y): \quad L(1.0(-\sin(\theta_5)\cos(\theta_6)\cos(\theta_7)+\sin(\theta_7)\cos(\theta_5))\cos(\theta_1)-1.0\sin(\theta_1)\sin(\theta_5)\sin(\theta_6)-2.0\text{s} \tag{115}$$

$$= -L(2.0\sin(\theta_3)+5.0\cos(\theta_3)) \tag{116}$$

$$[8pt](n_z): \quad \sin(\theta_1)\cos(\theta_6)-\sin(\theta_6)\cos(\theta_1)\cos(\theta_7) \tag{117}$$

$$= -\sin(\theta_2)\cos(\theta_4)+\sin(\theta_4)\cos(\theta_2)\cos(\theta_3) \tag{118}$$

$$[8pt](o_z): \quad \sin(\theta_1)\sin(\theta_6)\cos(\theta_7)+\cos(\theta_1)\cos(\theta_6) \tag{119}$$

$$= \sin(\theta_2)\sin(\theta_4)\cos(\theta_3)+\cos(\theta_2)\cos(\theta_4) \tag{120}$$

$$[8pt](a_z): \quad \sin(\theta_6)\sin(\theta_7) \tag{121}$$

$$= -\sin(\theta_3)\sin(\theta_4) \tag{122}$$

$$[8pt](p_z): \quad L(1.0\sin(\theta_1)\cos(\theta_6)-1.0\sin(\theta_6)\cos(\theta_1)\cos(\theta_7)-2.0\sin(\theta_6)\cos(\theta_7)) \tag{123}$$

$$= L(5.0\sin(\theta_3)-2.0\cos(\theta_3))\sin(\theta_4) \tag{124}$$

$$[8pt] \tag{125}$$

7 Permutation 6: $T_6T_7T_1T_2 = T_5^{-1}T_4^{-1}T_3^{-1}$

$$
\begin{aligned}
(n_x): &\quad -\sin(\theta_6)\sin(\theta_1+\theta_2) - \cos(\theta_6)\cos(\theta_7)\cos(\theta_1+\theta_2) && (126)\\
&= \sin(\theta_3)\sin(\theta_5) - \cos(\theta_3)\cos(\theta_4)\cos(\theta_5) && (127)\\
[8pt](o_x): &\quad -\sin(\theta_7)\cos(\theta_6) && (128)\\
&= -\sin(\theta_3)\cos(\theta_4)\cos(\theta_5) - \sin(\theta_5)\cos(\theta_3) && (129)\\
[8pt](a_x): &\quad -\sin(\theta_6)\cos(\theta_1+\theta_2) + \sin(\theta_1+\theta_2)\cos(\theta_6)\cos(\theta_7) && (130)\\
&= \sin(\theta_4)\cos(\theta_5) && (131)\\
[8pt](p_x): &\quad L(-1.0\sin(\theta_1)\sin(\theta_6) - 2.0\sin(\theta_6)\sin(\theta_1+\theta_2) + 5.0\sin(\theta_7)\cos(\theta_6) - 1.0\cos(\theta_1)\cos(\theta_6)\cos && (132)\\
&= -2.5L(\cos(\theta_5)+1) && (133)\\
[8pt](n_y): &\quad -\sin(\theta_6)\cos(\theta_7)\cos(\theta_1+\theta_2) + \sin(\theta_1+\theta_2)\cos(\theta_6) && (134)\\
&= \sin(\theta_4)\cos(\theta_3) && (135)\\
[8pt](o_y): &\quad -\sin(\theta_6)\sin(\theta_7) && (136)\\
&= \sin(\theta_3)\sin(\theta_4) && (137)\\
[8pt](a_y): &\quad \sin(\theta_6)\sin(\theta_1+\theta_2)\cos(\theta_7) + \cos(\theta_6)\cos(\theta_1+\theta_2) && (138)\\
&= \cos(\theta_4) && (139)\\
[8pt](p_y): &\quad L(1.0\sin(\theta_1)\cos(\theta_6) + 5.0\sin(\theta_6)\sin(\theta_7) - 1.0\sin(\theta_6)\cos(\theta_1)\cos(\theta_7) - 2.0\sin(\theta_6)\cos(\theta_7)\cos && (140)\\
&= 0 && (141)\\
[8pt](n_z): &\quad -\sin(\theta_7)\cos(\theta_1+\theta_2) && (142)\\
&= -\sin(\theta_3)\cos(\theta_5) - \sin(\theta_5)\cos(\theta_3)\cos(\theta_4) && (143)\\
[8pt](o_z): &\quad \cos(\theta_7) && (144)\\
&= -\sin(\theta_3)\sin(\theta_5)\cos(\theta_4) + \cos(\theta_3)\cos(\theta_5) && (145)\\
[8pt](a_z): &\quad \sin(\theta_7)\sin(\theta_1+\theta_2) && (146)\\
&= \sin(\theta_4)\sin(\theta_5) && (147)\\
[8pt](p_z): &\quad L(-1.0\sin(\theta_7)\cos(\theta_1) - 2.0\sin(\theta_7)\cos(\theta_1+\theta_2) - 2.0\sin(\theta_7) - 5.0\cos(\theta_7) + 2.5) && (148)\\
&= -2.5L\sin(\theta_5) && (149)\\
[8pt] && (150)
\end{aligned}
$$

8 Permutation 7: $T_7T_1T_2T_3 = T_6^{-1}T_5^{-1}T_4^{-1}$

$$
\begin{aligned}
(n_x): &\quad -\sin(\theta_3)\sin(\theta_7) - \cos(\theta_3)\cos(\theta_7)\cos(\theta_1+\theta_2) && (151)\\
&= \sin(\theta_4)\sin(\theta_6) - \cos(\theta_4)\cos(\theta_5)\cos(\theta_6) && (152)\\
[8pt](o_x): &\quad -\sin(\theta_1+\theta_2)\cos(\theta_7) && (153)\\
&= -\sin(\theta_4)\cos(\theta_5)\cos(\theta_6) - \sin(\theta_6)\cos(\theta_4) && (154)\\
[8pt](a_x): &\quad -\sin(\theta_3)\cos(\theta_7)\cos(\theta_1+\theta_2) + \sin(\theta_7)\cos(\theta_3) && (155)\\
&= \sin(\theta_5)\cos(\theta_6) && (156)\\
[8pt](p_x): &\quad L\left(-5.0\sin(\theta_7) + 1.0\cos(\theta_1)\cos(\theta_7) + 2.0\cos(\theta_7)\cos(\theta_1+\theta_2) + 2.0\cos(\theta_7)\right) && (157)\\
&= 2.5L\left(\cos(\theta_5)+1\right)\cos(\theta_6) && (158)\\
[8pt](n_y): &\quad \sin(\theta_3)\cos(\theta_7) - \sin(\theta_7)\cos(\theta_3)\cos(\theta_1+\theta_2) && (159)\\
&= -\sin(\theta_5)\cos(\theta_4) && (160)\\
[8pt](o_y): &\quad -\sin(\theta_7)\sin(\theta_1+\theta_2) && (161)\\
&= -\sin(\theta_4)\sin(\theta_5) && (162)\\
[8pt](a_y): &\quad -\sin(\theta_3)\sin(\theta_7)\cos(\theta_1+\theta_2) - \cos(\theta_3)\cos(\theta_7) && (163)\\
&= -\cos(\theta_5) && (164)\\
[8pt](p_y): &\quad L\left(1.0\sin(\theta_7)\cos(\theta_1) + 2.0\sin(\theta_7)\cos(\theta_1+\theta_2) + 2.0\sin(\theta_7) + 5.0\cos(\theta_7)\right) && (165)\\
&= 2.5L\left(\sin(\theta_5)+1\right) && (166)\\
[8pt](n_z): &\quad \sin(\theta_1+\theta_2)\cos(\theta_3) && (167)\\
&= \sin(\theta_4)\cos(\theta_6) + \sin(\theta_6)\cos(\theta_4)\cos(\theta_5) && (168)\\
[8pt](o_z): &\quad -\cos(\theta_1+\theta_2) && (169)\\
&= \sin(\theta_4)\sin(\theta_6)\cos(\theta_5) - \cos(\theta_4)\cos(\theta_6) && (170)\\
[8pt](a_z): &\quad \sin(\theta_3)\sin(\theta_1+\theta_2) && (171)\\
&= -\sin(\theta_5)\sin(\theta_6) && (172)\\
[8pt](p_z): &\quad -L\left(1.0\sin(\theta_1) + 2.0\sin(\theta_1+\theta_2)\right) && (173)\\
&= -2.5L\left(\cos(\theta_5)+1\right)\sin(\theta_6) && (174)\\
[8pt] & && (175)
\end{aligned}
$$